



Shrushranto Rajbongshi

Aspiring Software Developer

Roll No.: 524110026

Master's Of Computer Applications

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GitHub Profile

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EDUCATION

- NATIONAL INSTITUTE OF TECHNOLOGY, KURUKSHETRA** *Aug 2024 – Present*
MASTER OF COMPUTER APPLICATIONS (MCA) *CGPA: 8.2*
- CHANDIGARH UNIVERSITY** *Jun 2021 – Jun 2024*
BACHELOR OF COMPUTER APPLICATION (BCA) *CGPA: 7.5*

SKILLS

Programming: Java, Python, C

Machine Learning: TensorFlow, Keras, scikit-learn, OpenCV

Tools & Technologies: Git, Jupyter Notebook, VS Code, IntelliJ, Power BI

Databases: MySQL, SQL

PROJECTS

- SiLo AI — Indian Sign Language Recognition System** *Jan 2026 – Present*
 - Proposed a novel ISL recognition pipeline using raw RGB frames, eliminating pose-based preprocessing (OpenPose) to improve efficiency and scalability.
 - Designed a hybrid deep learning architecture combining **MobileNetV2** and **Bidirectional LSTM (temporal)** for video sequence modeling.
 - Achieved **3.5× speedup** in inference time (130ms → 37ms) while maintaining **89.22% accuracy**.
 - Constructed INCLUDE_50_Clean dataset after identifying inconsistencies in the original dataset distribution.
 - Evaluated using **5-fold cross-validation with strict video-level splits**, eliminating data leakage and ensuring robust generalization.
 - Focus: Efficient video understanding using lightweight deep learning architectures for real-world deployment.
 - Tech Stack:** Python, TensorFlow, Keras, OpenCV, NumPy, scikit-learn, CUDA
- Crime Rate Analysis using Python & Power BI (GitHub: crime-rate-analysis)** *Jan 2025 – Jun 2025*
 - Performed analysis of national crime datasets (2018–2021), focusing on Cybercrime and IPC trends.
 - Identified Identity Theft as the fastest-growing cybercrime during increased digital activity.
 - Observed correlation between IPC and Cybercrime cases indicating shared socioeconomic factors.
 - Built Power BI dashboards to visualize trends, pendency, and resolution efficiency.
 - Presented findings in departmental seminars for policy-level insights.
 - Tech Stack:** Python, Pandas, Power BI, Excel
- Retro-Arcadia – Multi-Game Desktop Arcade (GitHub: Retro-Arcadia)** *Jan 2025 – Jun 2025*
 - Developed a modular arcade platform integrating multiple classic games.
 - Built Tetris using Python and Pygame with full game logic and UI.
 - Designed a centralized launcher for seamless navigation between games.
 - Applied OOP principles for scalable and maintainable architecture.
 - Tech Stack:** Python, Pygame, OOP

RESEARCH & PUBLICATIONS

- ICCCI 2026 — Paper Submission (Springer LNCS)** *March 2026*
 - Title:** SiLo AI: Efficient Indian Sign Language Recognition Using Raw RGB Frames and Bidirectional LSTM
 - Authors:** Shrushranto Rajbongshi, Priyanshu Mudgal, Dr. Sarika Jain
 - Submitted to the 18th International Conference on Computational Collective Intelligence (ICCCI 2026).

POSITIONS OF RESPONSIBILITY

- Head of Management Committee, Mexperts Technical Society – NIT Kurukshetra** *Aug 2024 – Present*
 - Led the Management Committee to plan and execute technical events, coding contests, and workshops on campus.
 - Coordinated with team members and faculty to ensure smooth operations and high student engagement.
 - Implemented strategies to improve event organization, resource allocation, and overall team efficiency.

CERTIFICATIONS

- Workshop on Power Query and Power BI** *Mar 29, 2025*
Conducted by the Department of Computer Applications, NIT Kurukshetra
- Master Data Analytics with MS Excel** *Jan 22, 2025*
Organized by Mexperts Technical Society & Institution's Innovation Council, NIT Kurukshetra